

A smooth counterexample to the negative- p normalized-moment conjecture

Automated mathematics run `fa_banach_001`
Lane 3 — candidate for human review

17 August 2026

Status. Candidate full counterexample, likely valid. The construction disproves the negative-exponent extension proposed in Remark 2.2 of Jie Xiao’s *Optimal Monotonicity of L^p Integral of Conformal Invariant Green Function*, arXiv:0708.2644 [1].

1 Source conjecture

Theorem 2.1(i) of the source fixes $c > 0$, assumes that $X \geq 0$ on $(0, \infty)$ and

$$X'(t) \leq 0, \quad \frac{d}{dt}(e^{ct}X(t)) \leq 0,$$

and defines

$$Y_p(0) = - \int_0^\infty t^p dX(t).$$

For $0 \leq p_1 < p_2$, it proves

$$\frac{c^{p_2}Y_{p_2}(0)}{\Gamma(p_2+1)} \leq \frac{c^{p_1}Y_{p_1}(0)}{\Gamma(p_1+1)}. \tag{1}$$

Remark 2.2, on source-PDF page 6, conjectures that Theorem 2.1(i) remains valid for $-1 < p_1 < 0$, and explicitly discusses $-1 < p_1 < p_2 < 0$. The original passage is reproduced below.

Remark 2.2. A close look at (2.2)-(2.3)-(2.4) leads us to conjecture that Theorem 2.1(i) is still valid for $-1 < p_1 < 0$. This thought is also supported by the following analysis:

(i) Using the Cauchy-Schwarz inequality and (2.2), we find that for $-1 < p_1 < 0$,

$$Y_0(0) \leq (Y_{p_1}(0))^{\frac{1}{2}} (Y_{-p_1}(0))^{\frac{1}{2}} \leq (Y_{p_1}(0))^{\frac{1}{2}} (c^{p_1} \Gamma(1-p_1) Y(0))^{\frac{1}{2}},$$

thereby getting

$$(2.6) \quad Y_0(0) \leq c^{p_1} \Gamma(1-p_1) Y_{p_1}(0) = \left(\frac{\pi p_1}{\sin \pi p_1} \right) \left(\frac{c^{p_1}}{\Gamma(1+p_1)} \right) Y_{p_1}(0).$$

The estimate (2.6) and the Hölder inequality yield that for $-1 < p_1 < p_2 < 0$,

$$(2.7) \quad Y_{p_2}(0) \leq \min \left\{ \left(\frac{\pi p_1 c^{p_1}}{(\sin \pi p_1) \Gamma(1+p_1)} \right)^{1-\frac{p_2}{p_1}}, \left(\frac{\pi p_2 c^{p_2}}{(\sin \pi p_2) \Gamma(1+p_2)} \right)^{\frac{p_2}{p_1}-1} \right\} Y_{p_1}(0).$$

However, when $X(t) = e^{-ct} X(0)$, the equalities in (2.6) and (2.7) do not occur.

(ii) Noticing that

$$\lim_{p \rightarrow -1^+} \frac{\Gamma(1+p)}{(1+p)^{-1}} = 1; \quad 0 \leq - \int_1^\infty \frac{dX(r)}{r^{-p}} < \infty \text{ for } p \in (-1, 0); \quad cX(t) \leq -X'(t),$$

and that $(1+p)r^p dr$, as a measure on $[0, 1]$, converges weakly to the point mass at $r = 0$ as $p \rightarrow -1^+$, we achieve

$$(2.8) \quad \begin{aligned} \lim_{p \rightarrow -1^+} \frac{c^p Y_p(0)}{\Gamma(1+p)} &= - \lim_{p \rightarrow -1^+} \frac{c^p}{\Gamma(1+p)} \left(\int_0^1 r^p dX(r) + \int_1^\infty r^p dX(r) \right) \\ &= -c^{-1} \lim_{p \rightarrow -1^+} \frac{\int_0^1 X'(r) dr^{1+p}}{(1+p)\Gamma(1+p)} \\ &= -c^{-1} \lim_{t \rightarrow 0^+} X'(t) \geq X(0). \end{aligned}$$

Thus it suffices to find one admissible X , with $c = 1$, for which the normalized moment

$$M(p) := \frac{Y_p(0)}{\Gamma(p+1)}$$

is not nonincreasing on $(-1, 0)$.

2 Proof intuition

For the exponential extremizer $X(t) = e^{-t}$, every normalized moment is exactly one. We perturb it by making $e^t X(t)$ decrease smoothly only after time one. This preserves both differential hypotheses and avoids the singularity at the origin. Integration by parts then shows that every negative normalized moment is *strictly larger* than one. Nevertheless, the Gamma factor diverges as $p \downarrow -1$, forcing these normalized moments back to one at the left endpoint. A nonincreasing function cannot have that behavior: starting at the limiting value one, it cannot later rise strictly above one.

3 Construction and counterexample

Put $c = 1$, and define a continuously differentiable, nondecreasing function $G : [0, \infty) \rightarrow [0, 1/2]$ by

$$G(t) = \begin{cases} 0, & 0 \leq t \leq 1, \\ \frac{1}{2}(3(t-1)^2 - 2(t-1)^3), & 1 < t < 2, \\ \frac{1}{2}, & t \geq 2. \end{cases} \quad (2)$$

Set

$$X(t) = e^{-t}(1 - G(t)). \quad (3)$$

Theorem 1. *The function X in (3) is positive and C^1 , satisfies both hypotheses of Theorem 2.1, and has finite $Y_p(0)$ for every $p > -1$. There exist $-1 < p_1 < p_2 < 0$ for which*

$$\frac{Y_{p_2}(0)}{\Gamma(p_2 + 1)} > \frac{Y_{p_1}(0)}{\Gamma(p_1 + 1)}.$$

Consequently, the extension proposed in Remark 2.2 is false.

Proof. The cubic in (2) has derivative $3(t-1)(2-t)$ before multiplication by $1/2$. Hence $G \in C^1$, $G' \geq 0$, and $0 \leq G \leq 1/2$. It follows directly that

$$X'(t) = -e^{-t}(1 - G(t) + G'(t)) < 0, \quad \frac{d}{dt}(e^t X(t)) = -G'(t) \leq 0. \quad (4)$$

In particular, all the source hypotheses hold. Near zero, $-X'(t) = e^{-t}$, while at infinity $-X'(t) = \frac{1}{2}e^{-t}$, so $Y_p(0) = \int_0^\infty t^p(-X'(t)) dt$ is finite for every $p > -1$.

For such p , ordinary integration by parts is legitimate because G vanishes on $[0, 1]$ and is bounded. From (4),

$$\begin{aligned} Y_p(0) &= \int_0^\infty t^p e^{-t}(1 - G(t) + G'(t)) dt \\ &= \Gamma(p+1) - p \int_0^\infty G(t) t^{p-1} e^{-t} dt. \end{aligned} \quad (5)$$

Indeed, the boundary term in integrating the G' -term is zero, and its $G(t)t^p e^{-t}$ contribution cancels the preceding $-G$ -term.

Write

$$J(p) = \int_0^\infty G(t) t^{p-1} e^{-t} dt.$$

Because $G = 0$ on $[0, 1]$, the function J is finite and continuous even on a neighborhood of $[-1, 0]$. Moreover $J(p) > 0$. Thus (5) gives

$$M(p) = 1 + \frac{(-p)J(p)}{\Gamma(p+1)} > 1 \quad (-1 < p < 0). \quad (6)$$

On the other hand, $J(p) \rightarrow J(-1) < \infty$, while $\Gamma(p+1) \rightarrow \infty$, as $p \downarrow -1$. Therefore

$$\lim_{p \downarrow -1} M(p) = 1. \quad (7)$$

Fix any $p_2 \in (-1, 0)$, for instance $p_2 = -1/2$. By (6), $M(p_2) > 1$; by (7), there is a $p_1 \in (-1, p_2)$ sufficiently close to -1 such that $M(p_1) < M(p_2)$. This is the strict reverse of (1). $\square$

4 Robustness and verification

The cubic profile is only a convenient explicit choice. The same proof works for every nonzero C^1 nondecreasing G that vanishes on a neighborhood of zero, is eventually constant, and satisfies $0 \leq G < 1$. Thus the failure is a stable family of counterexamples rather than an isolated numerical accident.

The script `code/check_counterexample.py`, run with 80 decimal digits, evaluates the exact right-hand side of (5). It reports

$$M(-0.99) = 1.0002754937621309\dots, \quad M(-0.5) = 1.0106701090171454\dots,$$

and confirms the displayed strict reversal. These decimal values are only QA; the existence argument from (6)–(7) is the proof.

5 Novelty check, limitations, and review focus

A bounded search on 17 August 2026 checked the run’s registry, solution, attempt, and proof-gap indexes; arXiv queries using the paper id, author, “negative p ”, and normalized Green-function moments; OpenAlex records and title/keyword searches; and Crossref title/author searches. No later paper or prior packet answering this exact negative-exponent conjecture was found. OpenAlex listed no citing works for the source record. This supports, but does not prove, novelty; the search was not an exhaustive MathSciNet or zbMATH review.

The result addresses the one-dimensional conjecture of Remark 2.2 exactly. It does not by itself realize (3) as a Green-function distribution, so it does not disprove the later geometric conjectures in Remarks 3.1 and 4.1. Human review should focus on the sign and boundary terms in (5), and on whether “Theorem 2.1(i) is still valid” is intended to retain precisely the stated one-dimensional hypotheses when $p_1 < 0$.

References

- [1] Jie Xiao, *Optimal Monotonicity of L^p Integral of Conformal Invariant Green Function*, arXiv:0708.2644 (2007), especially Theorem 2.1 and Remark 2.2.